Beyond the 4 °C Alarm

What a Temperature Log Can and Cannot Tell About Milk Spoilage

A loss-budget study of cold-chain monitoring

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Abstract

A cold-chain monitor records temperature, but whether milk spoils also depends on which bacteria it carries, how many and how ready they are to grow, none of which a thermometer observes. This monograph asks how much of the error in predicting spoilage from a temperature record is caused by that unobserved biology, how much by the sensor, how much by the sampling interval and how much by the choice of decision rule.

The question is answered with an open Monte Carlo benchmark of 240 000 simulated deliveries. A Baranyi–Roberts growth model with a square-root temperature law stands in for real milk, and a closed-form Gompertz shelf-life accumulator, cheap enough to run on a microcontroller, tries to predict it from noisy, sampled records. Every rule is judged by its ability to rank spoiled above unspoiled trips, measured by the area under the receiver operating characteristic curve (AUC), with paired bootstrap comparisons on identical trips and an oracle that knows each trip's biology. Parameters are tied to published experiments and to the ComBase database. The model reproduces a published fluctuating-temperature experiment within 4 %, fits 50 ComBase records of natural spoilage flora, and predicts 53 held-out growth rates of *Listeria monocytogenes* with an accuracy factor of 1.37.

Unobserved biology costs 0.27–0.40 AUC, an order of magnitude more than anything an engineer controls. A platinum resistance sensor instead of a low-cost digital probe gains at most 0.02; hourly sampling is indistinguishable from minute-by-minute sampling in 11 of 12 settings; and a growth-model score improves on a plain degree-hour count by at most 0.04. A Monte Carlo probability is far better calibrated than a single alarm. The same pattern holds in five parameter sets, including a hazard case study for *Listeria*. For this problem, the accuracy of a temperature monitor is limited by biology, not by the thermometer, and effort is better spent on measuring the product than on measuring its temperature more precisely.

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Notation

Symbol	Meaning	Unit
T, $T(t)$	product temperature, temperature history	°C
$\mu(T)$	maximum specific growth rate, natural-log basis	h^{-1}
μ_{10}	maximum growth rate, decimal-log basis, $\mu / \ln 10$	h^{-1}
b	slope of the square-root law	$\mathrm{h}^{-1/2}\,\mathrm{K}^{-1}$
$T_{\min}$	notional minimum growth temperature	°C
N, N_0, $N_{\max}$	bacterial count, starting count, maximum population density	$\mathrm{CFU\,mL}^{-1}$
N_L	limit against which spoilage is judged	$\mathrm{CFU\,mL}^{-1}$
y	natural logarithm of the count, $\ln N$	–
q, q_0	physiological state of the Baranyi–Roberts model and its initial value	–
h_0	lag parameter, $\ln(1+1/q_0)$; equal to $\lambda\mu$ at constant temperature	–
λ	lag time	h
A, L	$\log_{10}(N_{\max}/N_0)$ and $\log_{10}(N_L/N_0)$	–
$\mathrm{SL}(T)$	shelf life at a constant temperature T	h
K	shelf-life budget, $\mathrm{SL}(T)\,\mu(T)$	–
CSL	cumulative shelf life used; the limit is reached at $\mathrm{CSL} = 1$	–
Δt	sampling interval	min, h
cv	coefficient of variation of b between strains	–
σ_T	standard deviation of $T_{\min}$ between trips	K
AUC	area under the receiver operating characteristic curve	–
B_f, A_f	bias and accuracy factors of a growth-rate model	–
ECE	expected calibration error	–

CHAPTER 1

Introduction

1.1 The problem

Milk is perishable in a precise way. The psychrotrophic bacteria that spoil pasteurised milk grow slowly at 4 °C and several times faster at 13 °C, and every hand-off between the dairy, the truck, the depot, the shop and the household refrigerator is an opportunity for the temperature to drift. Predictive microbiology turns such a temperature history into an estimate of growth: a primary model describes the count against time, a secondary model describes the growth rate against temperature, and a tertiary model applies both along a measured history [1–3].

Temperature loggers are now cheap enough to travel with every crate [4–6], and dynamic shelf-life labels, which replace a printed date by a prediction from the log, have been proposed for dairy and meat [7–9]. A monitor of this kind will be trusted to decide which crates are sold first, which are tested and which are discarded. It must therefore come with an honest account of how often it is wrong, and why.

This monograph answers one question: *given only a temperature record, how well can an inexpensive calculation tell whether milk has passed a limit, and what prevents it from doing better?*

1.2 Why the question matters

Post-pasteurisation contamination with psychrotrophic bacteria causes roughly half to three-fifths of fluid-milk spoilage in the United States [10], and Gram-negative psychrotrophs that enter after pasteurisation are the main reason high-temperature short-time milk spoils early [11]. Foodborne disease more broadly causes an estimated 600 million illnesses and 420 000 deaths a year, 175 000 of them in the WHO South-East Asia region [12]. These deaths are dominated by agents other than milk-borne bacteria and are quoted only as context; this work models growth, not disease.

The limits a monitor reports against are process-hygiene criteria. The Food Safety and Standards Authority of India (FSSAI) sets an aerobic plate count of 3×10^4 CFU mL^{-1} for pasteurised milk [13], and quality deteriorates noticeably near 10^6 CFU mL^{-1} [10, 14]; both are used here. A recent review of fresh dairy concludes that few studies predict the end of shelf life, that endpoint definitions are inconsistent, that few models are validated beyond their development data, and that probabilities are preferable to single verdicts [15]. Without a quantitative account of where accuracy is lost, two opposite mistakes are made: money is spent on precise thermometers that cannot improve the prediction, and alarms are believed that are wrong a large fraction of the time.

1.3 Approach

A real cold chain never reveals the truth for every carton, so the question cannot be answered from field data alone. The approach taken here is a controlled simulation in which the truth is always known. Thousands of deliveries are generated with realistic temperature excursions and realistic variation in the bacteria; a reference model decides which cartons truly passed the limit; and a monitor that sees only a sensed, sampled temperature record tries to tell them apart. Information is then removed one source at a time, from an oracle that knows everything to a low-cost probe read once an hour, and the accuracy lost at each step is measured on the same trips. The result is a *loss budget*: a single table that states how much each source of error costs. Every parameter is tied to a published study, a

public database or an explicit assumption, and the model is checked against real growth data before any result is read.

1.4 Research questions

RQ1. How much ranking accuracy is lost to biology that a temperature record cannot see?
RQ2. How much is lost to the sensor and to the sampling interval, and how much does a better sensor or faster sampling recover?
RQ3. Does a growth-model score rank trips better than simple rules such as degree-hours, time above a threshold or the maximum temperature?
RQ4. Is a probability better calibrated than a single alarm?

1.5 Structure of the monograph

Chapter 2 reviews the microbiology of milk spoilage, predictive microbiology and existing approaches to cold-chain monitoring. Chapter 3 sets out the growth models and the edge estimator. Chapter 4 describes the benchmark and Chapter 5 its statistical methods. Chapter 6 validates the model against published and ComBase data. Chapter 7 reports the results, Chapter 8 the *Listeria* case study, and Chapter 9 the design of a monitor and the software. Chapter 10 discusses the findings and their limits, and Chapter 11 concludes. The appendices list the parameters with their provenance, the data and the procedure for reproducing every result.

CHAPTER 2

Background

Summary Pasteurised milk spoils mainly through psychrotrophic bacteria introduced after pasteurisation. Their growth is described by primary and secondary models fitted under constant conditions and applied along changing temperature histories. Existing work quantifies how variable spoilage is and which unknown input matters most; none measures what is left to predict once the temperature has been logged.

2.1 Spoilage of pasteurised milk

Pasteurisation destroys vegetative pathogens and most spoilage organisms, so the bacteria that spoil pasteurised fluid milk are either those that survive heating, chiefly spore-formers, or those that enter afterwards from fillers, pipework and the plant environment [10, 11]. Post-pasteurisation contaminants are dominated by Gram-negative psychrotrophs, above all *Pseudomonas*, which grow at refrigeration temperature, produce heat-stable proteases and lipases, and cause bitterness, off-flavours and coagulation once counts approach 10^6–10^7 CFU mL^{-1} [10]. The starting load is usually very low, often between one and fifty cells per 100 mL [10], so the time to spoilage is long at a well-controlled 4 °C and highly sensitive to warm intervals.

2.2 Predictive microbiology

2.2.1 Primary models

At constant temperature a bacterial population passes through a lag phase, an exponential phase and a stationary phase. The modified Gompertz model [16, 17] describes this sigmoid curve with parameters that have a direct biological meaning (lag, maximum rate and asymptote) and has a closed-form inverse, which makes it convenient for computing a time to a limit. The Baranyi–Roberts model [18, 19] describes the same curve through a differential equation with a physiological state variable that measures how ready the cells are to grow; because it is a differential equation, it handles changing temperature naturally, and it has become the reference model for dynamic conditions.

2.2.2 Secondary models

The secondary model gives the growth rate as a function of temperature. Below the optimum the square-root law of Ratkowsky [20] is the most widely used for psychrotrophs; the cardinal-temperature model [21] describes the whole biokinetic range, and Arrhenius-type laws with a fixed activation energy are used in time–temperature integrators [22]. ComBase gathers tens of thousands of growth records from which such models can be fitted [23, 24]. For psychrotrophic *Pseudomonas* in milk, the square-root law was fitted to 68 growth curves and tested in industry [25, 26].

2.2.3 Tertiary models and dynamic conditions

Models fitted at constant temperature are applied along changing histories by integrating the growth rate [27]. Growth under fluctuating temperature has been measured directly for *P. fluorescens* in milk [14], and shifts of temperature during the lag phase can add an extra lag [28]. Whether the extra lag matters for a monitor is one of the structural questions tested here.

2.3 Cold-chain monitoring

Time–temperature indicators and mean kinetic temperature integrate an Arrhenius rate with a fixed activation energy [9, 22]. The Safety Monitoring and Assurance System predicts remaining shelf life from logged temperature with a square-root model [7, 29]. Field studies of Swedish chains show large gaps between dynamically predicted and printed shelf life [8]; reviews describe how often cold chains break [30]; neural networks estimate product temperature from vehicle and door data [31]; heat-transfer simulation quantifies the cost of failures [32]; and recent work couples growth models with learned components [33].

2.4 Uncertainty analyses of shelf life

The studies closest to this work are Monte Carlo analyses of shelf life [34]. For fluid milk, Lau et al. modelled spoilage from Gram-negative post-pasteurisation contamination and found the maximum growth rate and the storage temperature most influential [35]. Buehler et al. and Qian et al. modelled psychrotolerant spore-formers and deployed the model as a supply-chain tool, with home storage temperature, transport temperature and initial spore load ranked highest [36, 37]. Giannakourou and Taoukis propagated kinetic uncertainty and cold-chain variability through nested Monte Carlo loops [38]. In all of these, temperature enters as an uncertain input drawn from a distribution.

2.5 The gap addressed here

Table 2.1 sets the approaches side by side. They ask how variable the outcome is, which input moves it most, or how to build a decision system. None asks the question a monitor faces: starting from a *measured* record and judged against a known truth, how much of the ability to rank trips by spoilage is lost to unobserved biology, to sensor error, to the sampling interval and to the choice of rule. The two views are complementary. When temperature is unknown it dominates the spread of outcomes; once it is logged, what remains is mostly biology, and that is what a loss budget measures.

Table 2.1 Approaches to predicting spoilage from a temperature history and the question each leaves open for a monitor

Approach	Input	Temperature law	Biological variability	Open question for a monitor
Threshold alarm, time above a limit	logged temperature	step	ignored	how often the alarm is wrong
Degree-hours	logged temperature	linear	ignored	how close it comes to a growth model
TTI, mean kinetic temperature [9, 22]	logged or physical integral	Arrhenius	ignored	accuracy lost to biology
Dynamic shelf life, SMAS [7, 29]	logged temperature	square-root	nominal	loss to biology, sensor and sampling
Monte Carlo spoilage models [35–37]	simulated temperature	square-root	sampled	what remains once temperature is logged
Nested uncertainty [38]	simulated temperature	Arrhenius	sampled	accuracy of a monitor
Product-temperature estimation [31]	vehicle data	Q_{10}	ignored	effect on ranking of spoiled units
This work	**sensed, sampled log**	**square-root**	**sampled, with oracle**	–

CHAPTER 3

Models

Summary A square-root law gives the growth rate; a Baranyi–Roberts model stands in for real milk; and a Gompertz accumulator, deliberately a different model, is what the monitor computes. The accumulator reduces a temperature record to the integral of the growth rate divided by a fixed budget, which is why a log can be read cheaply and why the budget, which carries the biology, limits accuracy.

3.1 Temperature and growth rate

The square-root law relates the maximum specific growth rate μ (h^{-1}, natural-log basis) to temperature below the optimum [20],

$$\sqrt{\mu(T)} = b\,(T - T_{\min}), \qquad T > T_{\min}, \tag{3.1}$$

and $\mu = 0$ at or below $T_{\min}$. Neumeyer fitted Eq. (3.1) to 68 growth curves of a fast-growing *Pseudomonas* in broth and milk, giving $T_{\min} = -7.7\,^{\circ}\mathrm{C}$ and, on the natural-log basis used here, $b = 0.030\,\mathrm{h}^{-1/2}\,\mathrm{K}^{-1}$ [25]. Rates of *P. fluorescens* in UHT milk at 4, 15 and 29 °C [14] give $b = 0.027$ and $T_{\min} = -9.0\,^{\circ}\mathrm{C}$. The two independent sources agree within 15 % on the generation time at 4 °C, 5.6 and 6.4 h.

The law implies that the rate grows with the square of the distance from $T_{\min}$:

$$\frac{\mu(T_2)}{\mu(T_1)} = \left(\frac{T_2 - T_{\min}}{T_1 - T_{\min}}\right)^2 . \tag{3.2}$$

With $T_{\min} = -7.7\,^{\circ}\mathrm{C}$, one hour at 13 °C is worth $(20.7/11.7)^2 = 3.1$ hours at 4 °C, and one hour at 17.5 °C is worth 4.6 hours. Short warm spells are therefore expensive, and the cost rises faster than the temperature.

3.2 The reference model

The Baranyi–Roberts model [18, 19] stands in for real milk. For $y = \ln N$ and the physiological state q,

$$\frac{\mathrm{d}y}{\mathrm{d}t} = \mu(T)\,\frac{q}{1+q}\left(1 - \mathrm{e}^{\,y - y_{\max}}\right), \tag{3.3}$$

$$\frac{\mathrm{d}q}{\mathrm{d}t} = \mu(T)\,q, \qquad h_0 = \ln(1 + 1/q_0)\,. \tag{3.4}$$

The factor $q/(1+q)$ starts small and rises towards one as the cells adapt, which produces the lag; the factor $1 - \mathrm{e}^{\,y - y_{\max}}$ stops growth at the maximum density. At constant temperature the lag is $\lambda = h_0/\mu$, so the product $\lambda\mu = h_0$ is constant for cells with the same history: cells that are slow to start are slow at every temperature in proportion. Because μ is evaluated along the trace, Eqs. (3.3)–(3.4) apply to any temperature history. They are integrated on a one-minute grid.

3.3 The edge estimator

A logger cannot integrate differential equations for many parameter sets cheaply, and the estimator should in any case differ from the reference so that the comparison is not circular. The monitor therefore uses the modified Gompertz curve [17],

$$\log_{10}\frac{N(t)}{N_0} = A\exp\Big\{-\exp\Big[\frac{\mu_{10}\,\mathrm{e}}{A}(\lambda - t)+1\Big]\Big\}, \tag{3.5}$$

with $\mu_{10} = \mu/\ln 10$ and $A = \log_{10}(N_{\max}/N_0)$. Setting $\log_{10}(N/N_0) = L = \log_{10}(N_L/N_0)$ and solving for t gives the shelf life at a constant temperature,

$$\mathrm{SL}(T) = \lambda + \frac{A}{\mu_{10}(T)\,\mathrm{e}}\left[1 - \ln\left(-\ln\frac{L}{A}\right)\right], \qquad \lambda = \frac{h_0}{\mu(T)}. \tag{3.6}$$

Both terms are proportional to $1/\mu(T)$, so

$$\mathrm{SL}(T) = \frac{K}{\mu(T)}, \qquad K = h_0 + \frac{A\ln 10}{\mathrm{e}}\left[1 - \ln\left(-\ln\frac{L}{A}\right)\right], \tag{3.7}$$

where the budget K depends on the lag and on the starting, limiting and maximum counts, but not on temperature.

3.4 Cumulative shelf life

A reading T_i held over an interval Δt_i consumes a fraction $\Delta t_i/\mathrm{SL}(T_i)$ of the shelf life. The cumulative shelf life used is

$$\mathrm{CSL} = \sum_i \frac{\Delta t_i}{\mathrm{SL}(T_i)}, \qquad \text{limit reached when } \mathrm{CSL} \geq 1, \tag{3.8}$$

and, by Eq. (3.7),

$$\mathrm{CSL} = \frac{1}{K}\sum_i \mu(T_i)\,\Delta t_i \;\approx\; \frac{1}{K}\int_0^t \mu\big(T(s)\big)\,\mathrm{d}s. \tag{3.9}$$

The monitor integrates the growth rate along the log and compares the result with a fixed budget. In practice $\mathrm{SL}(T)$ is stored as a table on a 0.01 °C grid, and each reading costs one table look-up and one addition.

Equation (3.9) separates what the record can supply from what it cannot. The integral is a property of the temperature history and is measured by the logger. The budget K and the rate scale b^2 are properties of the bacteria in the carton: their starting load, their physiological state and their strain. No temperature record contains them.

3.5 Degree-hours as an approximation

Writing $T - T_{\min} = (T_r - T_{\min}) + (T - T_r)$ in Eq. (3.1) and expanding the square shows that the integral in Eq. (3.9) consists of a term common to all trips of equal length, a term proportional to the signed degree-hours about T_r, and a quadratic term that gives extra weight to large excursions. Near refrigeration temperature the linear term dominates: for $T_r = 4$ °C, an excursion to 13 °C receives 211 units from the linear term and 81 from the quadratic term, so degree-hours capture about 72 % of its cost, and about 47 % for an excursion to 30.7 °C. A degree-hour count should therefore rank trips almost as well as the growth-model score, losing a little on hot excursions. Chapter 7 tests this.

3.6 A worked example

With the baseline parameters, $\mu(4) = (0.030 \times 11.7)^2 = 0.123\,\mathrm{h}^{-1}$ and $\mu(13) = 0.386\,\mathrm{h}^{-1}$. For the 3×10^4 CFU mL^{-1} limit the shelf life is about 83 h at a steady 4 °C and about 26 h at 13 °C. One hour at 4 °C consumes 1.2 % of the shelf life and one hour at 13 °C 3.8 %, so a 90-min spell at 13 °C costs about 4 % of shelf life beyond what the same time at 4 °C would have cost. Figure 3.1 follows a single excursion through the reference model and the accumulator.

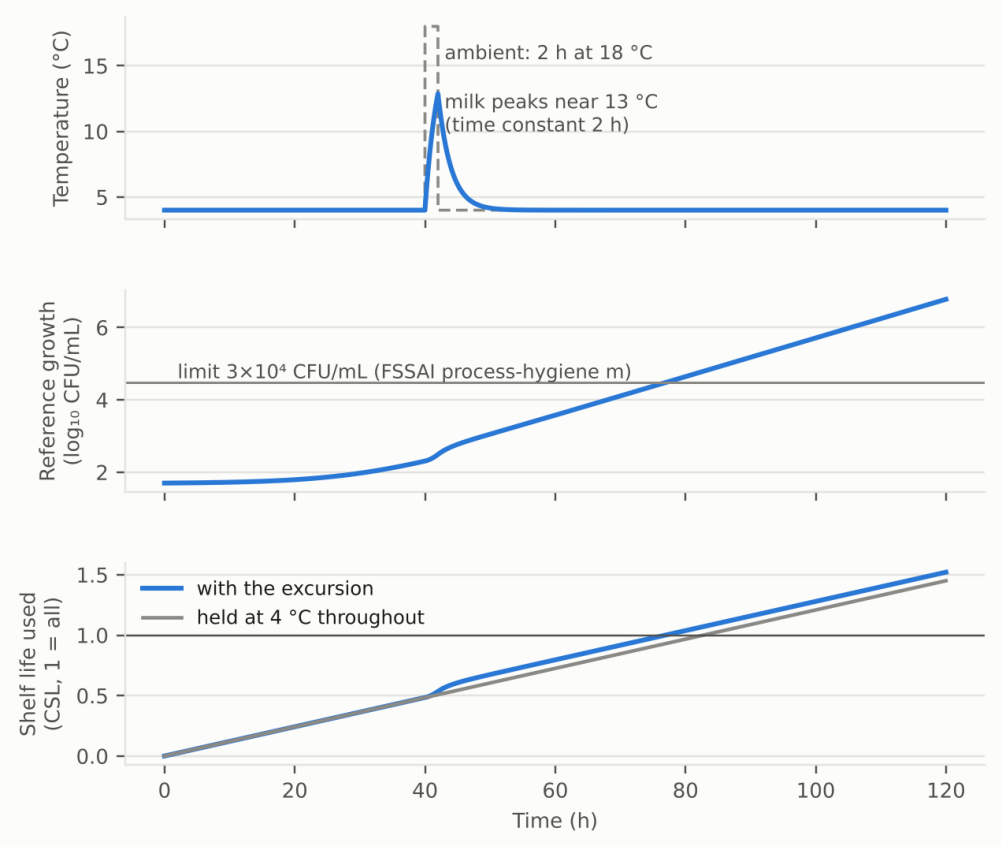


Figure 3.1 One excursion through the models. Top: ambient temperature (2 h at 18 °C) and product temperature of an insulated crate. Middle: reference growth. Bottom: cumulative shelf life with and without the excursion

3.7 An illustrative trip

Figure 3.2 shows the central difficulty on one 72-h delivery. Three cartons travel together and therefore share a single temperature history; they differ only in their starting load and the physiological state of their bacteria. The monitor, reading the log, returns one prediction for all three. It is late for one carton and early for the other two, and a more accurate thermometer cannot change that, because the information it lacks is not in the temperature record.

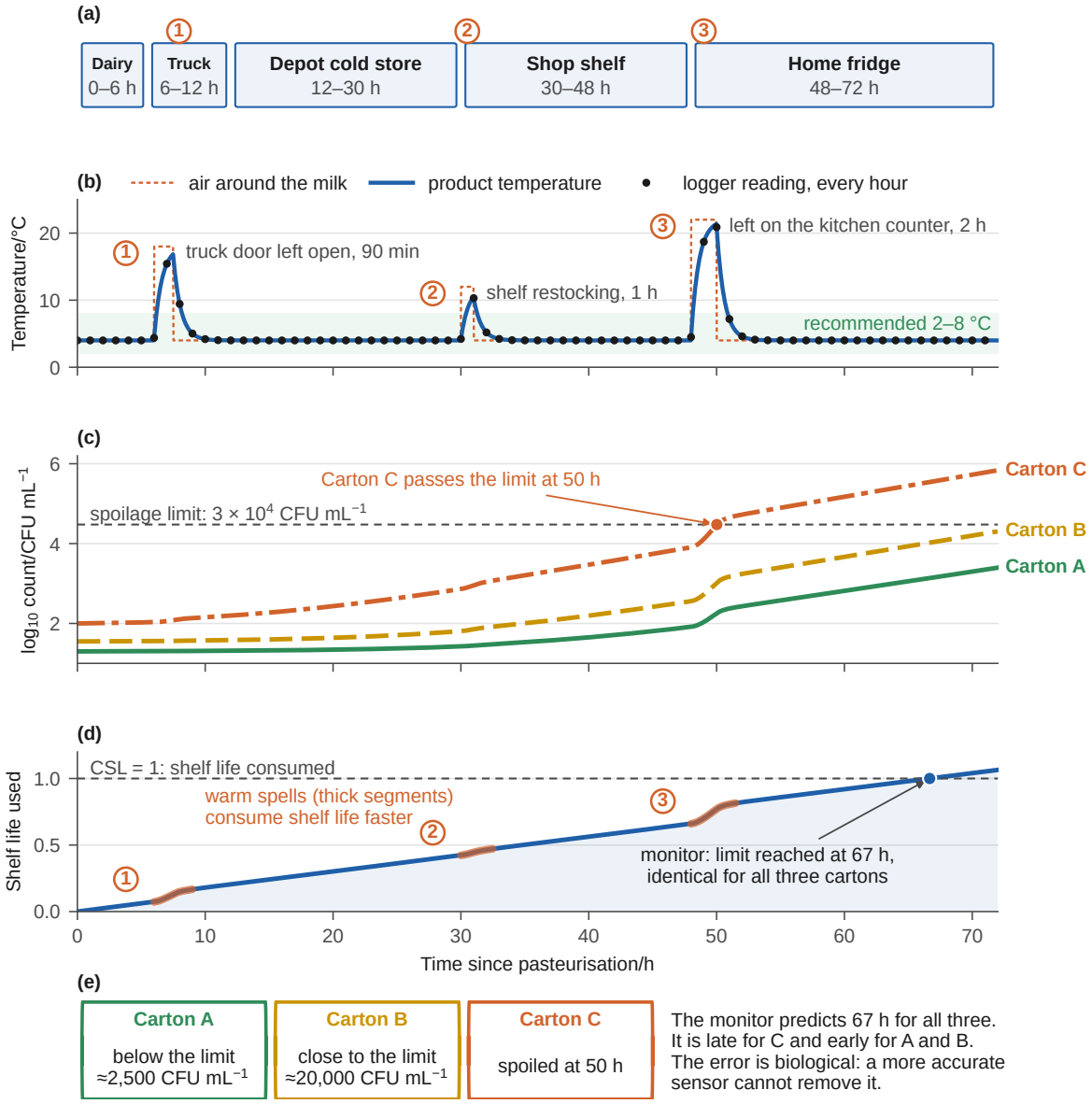


Figure 3.2 One delivery, three cartons. (a) Stages of the trip. (b) Ambient air and product temperature with three excursions and hourly logger readings. (c) Simulated counts in three cartons with the same history but different starting load and lag. (d) Cumulative shelf life computed by the monitor; warm spells consume shelf life faster. (e) One prediction for three cartons whose true fates differ

CHAPTER 4

The benchmark

Summary Each simulated trip has a true temperature history and a true biology. The reference model labels it spoiled or not; the monitor sees only a sensed, sampled record. Five parameter sets, 12 settings each and 4 000 trips per setting give 240 000 trips, each read with every sensor, sampling rule and score.

4.1 Design

Figure 4.1 summarises the design. A simulated world supplies the truth: a temperature history, a biology and, through the reference model, a label. The monitor sees only a record that has passed through a sensor and a sampling rule, and turns it into a score. Scores are compared with labels by AUC. An oracle with the true biology and the true temperature sets the ceiling, and removing information one step at a time yields the loss budget.

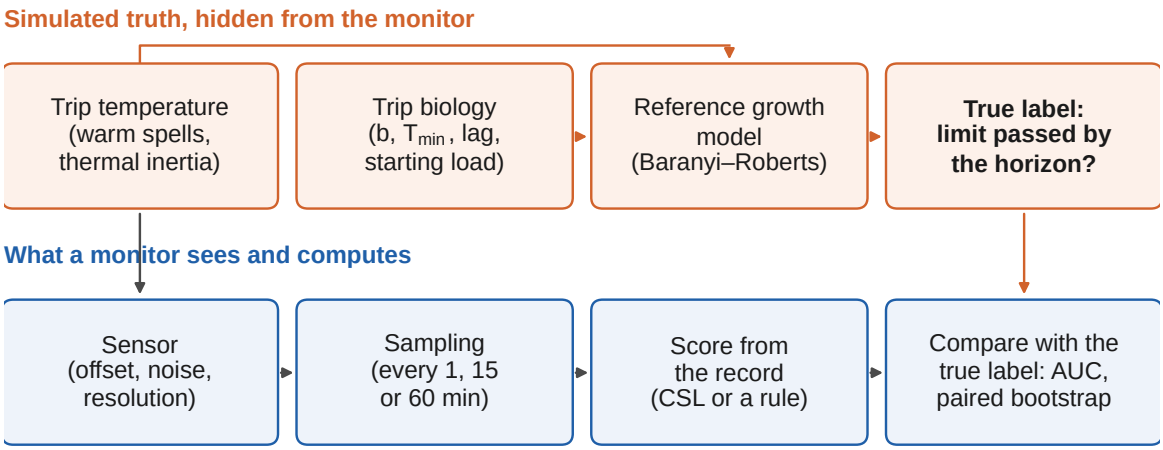


Figure 4.1 Design of the benchmark. Top: simulated truth, hidden from the monitor. Bottom: what the monitor sees and computes. Scores are judged against the true labels by AUC

4.2 Trip generator

Each trip has a product-temperature trace on a one-minute grid. The base temperature of the chain is drawn uniformly from 2.5–5 °C. Excursions arrive as a Poisson process with a mean of one per 144 h of horizon (at most eight per trip); each peaks at 12, 14.3, 17.5 or 30.7 °C with weights 0.30, 0.25, 0.30 and 0.15, representing transport and retail handling, and lasts a lognormal time with median 60 min and log-standard deviation 0.8, clipped to 10–480 min. Product temperature follows the ambient temperature with first-order dynamics,

$$\frac{\mathrm{d}T}{\mathrm{d}t} = \frac{T_a(t) - T}{\tau_p}, \tag{4.1}$$

with the product time constant τ_p drawn from 5–20 min for a low-mass pouch and 1–4 h for an insulated crate. The two products form one axis of the design: in a pouch the milk follows the air closely, while in a crate a short excursion is strongly damped.

4.3 Reference biology

For each trip the reference population starts from a random load and grows by Eqs. (3.3)–(3.4) along the trace, with

- $T_{\min} \sim \mathcal{N}(\bar{T}_{\min}, \sigma_T^2)$, the spread between strains and trips;
- $b = \bar{b}\,\varepsilon$ with ε lognormal with coefficient of variation $cv \in \{0, 0.10, 0.25\}$;
- h_0 uniform over a range derived from measured lag times;
- $\log_{10} N_0$ uniform over an assumed range.

In the structure-risk run, a quarter of the lag state is re-imposed whenever the product temperature departs from, or returns to, its two-hour running mean by more than 4 °C, following the extra lag observed after temperature shifts [28]. The estimator is not told about this mechanism.

4.4 Labels

A trip is *spoiled* if the reference population exceeds the limit within the label horizon. The horizon is the time the estimator takes to reach the limit at a constant 4 °C with nominal parameters, rounded up to a whole hour, as a printed shelf life would be. It was fixed from label balance before any rule was compared. Prevalence is therefore a property of the design (about 0.5 for spoilage and 0.08–0.38 for *Listeria*), and absolute predictive values must not be read as field rates. Only ranking (AUC) and calibration against the constructed labels are interpreted.

4.5 Sensors and sampling

Each sensor adds a per-trip offset drawn uniformly within its accuracy band, white noise and quantisation:

- DS18B20 digital sensor: offset within ±0.5 °C [39], noise 0.05 °C, resolution 0.0625 °C;
- PT1000 platinum resistance sensor read by a MAX31865: offset within ±0.15 °C (IEC 60751 Class A [40]), noise 0.03 °C, resolution 0.03125 °C;
- an ideal sensor with no error.

Sampling intervals are 1, 15 and 60 min, plus an adaptive rule that samples every minute while the temperature is changing and every 15 min otherwise. Every trip is read with every sensor and sampling rule, so that differences between them are measured on identical histories.

4.6 Scores

Five scores are computed from the measured record: the cumulative shelf life CSL of Eq. (3.8) with nominal parameters; degree-hours above 4 °C; time above 5 °C; time above 8 °C; and the maximum temperature. The oracle computes CSL with each trip's true parameters and noise-free temperature. A sixth output, a probability, is described in Sect. 4.8.

4.7 Parameter sets

Table 4.1 lists the five runs. The baseline uses the Neumeyer parameters and a measured lag range. The extra-lag run adds the structural risk described above. The low-load run uses a starting load close to real post-pasteurisation levels. Because parameters from inoculated isolates need not describe natural flora, the milk-flora run uses parameters fitted to ComBase records of spoilage bacteria in pasteurised milk (Chapter 6). The *Listeria* run models a pathogen with labels of $+0.5$ and $+1$ $\log_{10}$ growth above the starting load; these are round-number labels, not regulatory limits. Each run has 12 settings (two limits, two products, three values of *cv*) of 4 000 trips. The full list of parameters, with the source of every value, is given in Appendix A.

Table 4.1 Benchmark runs. “Assumed” marks values without a direct source

Run	Biology and labels
Baseline	$b = 0.030$, $T_{\min} = -7.7\,^{\circ}\mathrm{C}$ [25]; $\sigma_T = 1.5\,\mathrm{K}$ (assumed); $h_0 \in [2.5, 5]$ [14]; N_0 30–100 CFU mL^{-1} (assumed); limits 3×10^4 and 10^6 CFU mL^{-1}; horizons 83 and 113 h
Extra lag	as baseline, plus extra lag after temperature shifts [28]
Low load	as baseline, N_0 0.1–3 CFU mL^{-1} [10]; horizons 120 and 154 h
Milk flora	$b = 0.0256$, $T_{\min} = -6.6\,^{\circ}\mathrm{C}$ (ComBase, $n = 50$), low load; horizons 201 and 258 h
Listeria	$b = 0.0235$, $T_{\min} = -2.3\,^{\circ}\mathrm{C}$ (ComBase, $n = 46$); $h_0 \in [2, 4.5]$, $\sigma_T = 1.0\,\mathrm{K}$, N_0 0.06–0.18 CFU mL^{-1} (assumed); labels +0.5 and +1 log; horizons 100 and 202 h

4.8 Probabilistic output

Instead of a single verdict, the monitor can draw $M = 80$ parameter sets $\theta_k = (b_k, T_{\min,k}, h_{0,k}, N_{0,k})$ from a prior and report

$$\hat{p} = \frac{1}{M} \sum_{k=1}^{M} \mathbf{1}\{\mathrm{CSL}(\theta_k) \geq 1\}, \tag{4.2}$$

the fraction of plausible biologies under which the limit has been reached. Its Monte Carlo standard error is at most $\sqrt{0.25/80} = 0.056$. Two priors are compared with the single alarm that uses nominal parameters: a prior that matches the simulated variation, and an overconfident prior that ignores the strain variation in b.

4.9 Protocol

For every run and setting: (1) generate 4 000 temperature histories; (2) simulate the reference population and label each trip; (3) read each history with every sensor and sampling rule; (4) compute all scores and the oracle; (5) estimate every AUC with a bootstrap interval and every paired contrast with a paired-bootstrap interval; (6) compute the probability and its calibration. Random seeds are fixed, and the whole benchmark runs in about three hours on two processor cores.

CHAPTER 5

Statistical methods

Summary Rules are compared by AUC, which measures ranking and does not depend on the constructed prevalence. Differences between rules are tested on identical trips with a paired bootstrap, and calibration is measured by the Brier score and the expected calibration error.

5.1 The area under the ROC curve

Let S_1 be the score of a randomly chosen spoiled trip and S_0 that of a randomly chosen unspoiled trip. The primary metric is

$$\mathrm{AUC} = \Pr(S_1 > S_0) + \tfrac{1}{2}\Pr(S_1 = S_0), \tag{5.1}$$

the probability that the monitor ranks a spoiled trip above an unspoiled one [41]. A value of 0.5 is a coin toss and 1 is perfect ranking. AUC does not depend on a threshold or on the prevalence, which in this benchmark is constructed, so it measures how much information a score carries. For n_1 spoiled and n_0 unspoiled trips it is estimated by the Mann–Whitney statistic [42],

$$\widehat{\mathrm{AUC}} = \frac{1}{n_1 n_0}\Big(\sum_{i:y_i=1} r_i - \frac{n_1(n_1+1)}{2} \Big), \tag{5.2}$$

where r_i are the ranks of the pooled scores.

5.2 Precision and sample size

Hanley and McNeil approximated the standard error of $\widehat{\mathrm{AUC}}$ by

$$\mathrm{SE}^2 = \frac{A(1-A) + (n_1-1)(Q_1 - A^2) + (n_0-1)(Q_2 - A^2)}{n_1 n_0}, \tag{5.3}$$

with $Q_1 = A/(2-A)$ and $Q_2 = 2A^2/(1+A)$ [41]. For $A = 0.67$ and $n_1 = n_0 = 2\,000$, a typical setting, $\mathrm{SE} = 0.0085$. This fixed the number of trips: the loss to biology is more than fifteen standard errors, while sensor and sampling effects are of the order of one standard error and can be resolved only by paired comparison.

5.3 Bootstrap intervals

Intervals for single AUCs are percentile intervals from $B = 200$ bootstrap resamples of trips [43]. The trip is the resampling unit because trips are independent by construction, while the many readings within a trip are not.

5.4 Paired comparisons

Two scores computed on the same trips are correlated, because both respond to the same temperature histories. For a difference $D = \mathrm{AUC}_1 - \mathrm{AUC}_2$,

$$\operatorname{Var}(\hat{D}) = \operatorname{Var}(\widehat{\mathrm{AUC}}_1) + \operatorname{Var}(\widehat{\mathrm{AUC}}_2) - 2\operatorname{Cov}(\widehat{\mathrm{AUC}}_1, \widehat{\mathrm{AUC}}_2), \tag{5.4}$$

and the covariance term is large and positive for similar scores. Testing the two AUCs separately would ignore it and lose most of the power; this is why every trip is read with every sensor, sampling rule and score. The interval for D is obtained by resampling the same trips for both scores $B = 400$ times, the resampling counterpart of the method of DeLong et al. [44].

5.5 Multiplicity

Each run has 12 settings and each contrast is examined in all of them. No conclusion is drawn from a single setting: for each contrast the range of the difference over the settings is reported, together with the number of settings whose 95 % interval excludes zero. Under no real effect about 0.3 such exclusions in a favourable direction would be expected among 12 settings by chance.

5.6 Calibration

A probability $\hat{p}_i$ is calibrated if, among trips given probability p, a fraction p is spoiled. Calibration is measured by the Brier score and by the expected calibration error over ten equal-width bins [45, 46],

$$\mathrm{Brier} = \frac{1}{n}\sum_{i}(\hat{p}_i - y_i)^2, \tag{5.5}$$

$$\mathrm{ECE} = \sum_{m=1}^{10} \frac{|B_m|}{n}\left|\bar{y}_{B_m} - \bar{p}_{B_m}\right|. \tag{5.6}$$

5.7 Growth-rate fits

Secondary models are fitted by ordinary least squares on $\sqrt{\mu}$, with 95 % intervals from bootstrap resamples of records. Agreement between predicted and observed rates is summarised by the bias and accuracy factors of Ross [47],

$$B_f = 10^{\frac{1}{n}\sum \log_{10}(\hat{\mu}_i/\mu_i)}, \qquad A_f = 10^{\frac{1}{n}\sum |\log_{10}(\hat{\mu}_i/\mu_i)|}, \tag{5.7}$$

where $B_f = 1$ means no bias and $A_f - 1$ is the typical relative error.

CHAPTER 6

Validation against real data

Summary The model reproduces a published fluctuating-temperature experiment within 4 %, fits natural spoilage flora with the same temperature shape as inoculated isolates but a lower level, and predicts held-out *Listeria* growth rates without bias.

6.1 A fluctuating-temperature experiment

Lin et al. held UHT milk inoculated with *P. fluorescens* at 4 °C, with and without two four-hour excursions to 29 °C, and report that the time to 10^6 CFU mL^{-1} fell from about 90 h to about 55 h [14]. With parameters fitted only to their growth rates and their 4 °C lag, the reference model gives 90 h and 53 h (Fig. 6.1), an error of −4 % on the excursion case, which was not used for fitting. This is one experiment with one organism and an inoculated load far above real post-pasteurisation levels, and the published times are approximate values from the text.

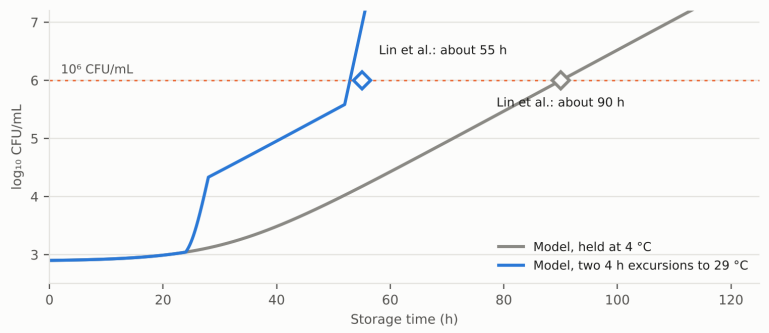


Figure 6.1 Reproduction of the experiment of Lin et al. [14]: model growth at a constant 4 °C and with two four-hour excursions to 29 °C, with the published times to 10^6 CFU mL^{-1} (diamonds)

6.2 Natural spoilage flora

ComBase holds 51 records of the maximum growth rate of aerobic spoilage bacteria in pasteurised milk at 2.7–14.9 °C from the Tasmanian Institute of Agriculture [24]; they are listed in Appendix B. One record at 6.8 °C lies far below its neighbours. Without it, the square-root law fits the remaining 50 records with $b = 0.0256$, $T_{\min} = -6.6$ °C and a residual standard deviation of 0.16 in natural-log units; with it, $b = 0.0260$ and $T_{\min} = -6.3$ °C, so no conclusion depends on it. Parameters from inoculated isolates overpredict these records with a bias factor of 1.62 that is nearly constant across temperature (Fig. 6.2): the temperature *shape* transfers from isolates to natural flora, the *level* does not. This is consistent with studies that choose a fast isolate as a worst case, and it is the reason for the milk-flora run.

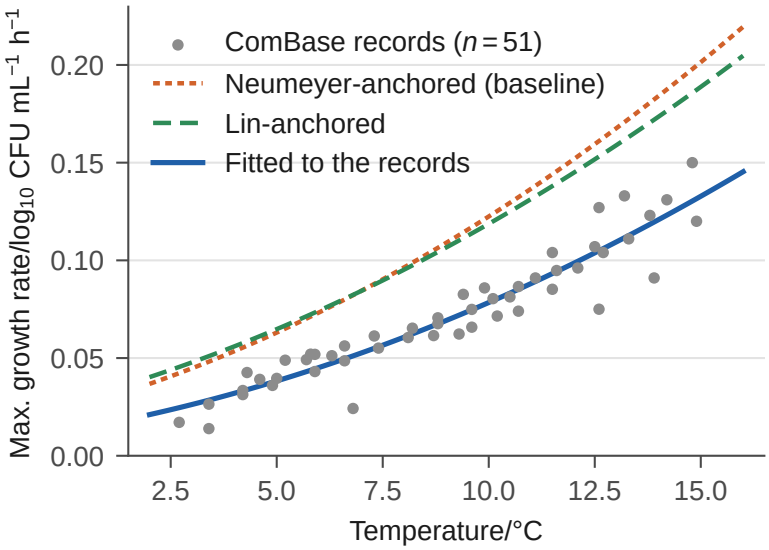


Figure 6.2 Maximum growth rates of natural spoilage flora in pasteurised milk (points; ComBase [24]) with the Neumeyer-anchored model (dotted), the Lin-anchored model (dashed) and the square-root fit to the records (solid)

6.3 *Listeria monocytogenes*

ComBase milk records with a stated growth rate for *L. monocytogenes* at 2–16 °C were split by condition: 46 records without added factors were used for fitting and 53 records with other stated conditions (sterilised or previously heated milk, added salts or acids) were held out. The fit gives $b = 0.0235$ (95 % bootstrap interval 0.0203–0.0265) and $T_{\min} = -2.3$ °C (−4.2 to −0.9 °C), a generation time of 32 h at 4 °C, and on the held-out records $B_f = 1.02$ and $A_f = 1.37$ ($A_f = 1.33$ on the fitted records). The split is by condition rather than by study, so some held-out records share a source with fitted ones and the test is weaker than a split by source. Because $T_{\min}$ lies closer to refrigeration temperature, the rate at 13 °C is 5.9 times that at 4 °C for *Listeria*, against about 3.4 times for spoilage flora (Fig. 6.3).

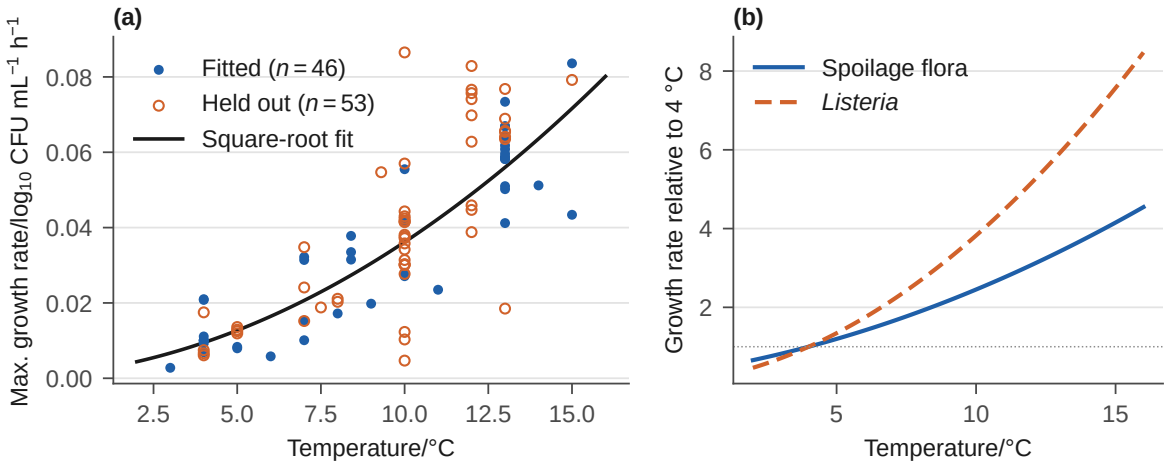


Figure 6.3 *Listeria monocytogenes* in milk. (a) Maximum growth rates used for fitting (filled) and held out by condition (open), with the square-root fit (ComBase [24]). (b) Growth rate relative to 4 °C for spoilage flora (solid) and *Listeria* (dashed)

Table 6.1 Composition of the compiled ComBase milk database by organism. *n*: records; Rate: maximum growth rate stated; Curve: growth curve only; None: no growth reported; N/T: rate not transcribed

Organism	*n*	Rate	Curve	None	N/T
L. monocytogenes/innocua	149	102	43	0	4
Bacillus cereus	94	47	36	3	8
Escherichia coli	66	22	15	4	25
Salmonella spp.	25	4	7	13	1
Campylobacteraceae	16	0	13	0	3
Shigellae	11	0	10	0	1
Cronobacter spp.	6	0	5	0	1
Aerobic spoilage bacteria	4	4	0	0	0
Lactic acid bacteria	1	0	1	0	0
Other	1	0	1	0	0
Total	373	179	131	20	43

6.4 The compiled data

A ComBase search with milk as the food matrix returned about 480 records, of which 373 were compiled (Table 6.1), together with the 51 records of spoilage flora. Records without a stated rate were not used for fitting. All compiled records are released with the code.

CHAPTER 7

Results

Principal findings

- Unobserved biology costs 0.27–0.40 AUC in the baseline run, an order of magnitude more than any sensor or sampling choice.
- A PT1000 instead of a DS18B20 gains at most 0.02 AUC; hourly sampling matches one-minute sampling in 11 of 12 settings.
- The growth-model score beats time above 5 °C and the maximum temperature in every setting, but improves on degree-hours by at most 0.04.
- A Monte Carlo probability reaches ECE 0.03–0.07; a single alarm 0.34–0.43.
- The pattern is the same in all five parameter sets.

7.1 Loss budget (RQ1)

An oracle that knows each trip's biology and temperature separates spoiled from unspoiled trips almost perfectly, by construction. The same CSL with nominal biology and an ideal sensor loses 0.27–0.40 AUC across the 12 baseline settings (Table 7.1, Fig. 7.1). The loss is large even when strains do not differ in growth rate ($cv = 0$), because the minimum growth temperature, the lag state and the starting load still differ between trips. In terms of Eq. (3.9), these quantities set the rate scale and the budget K, which the temperature record does not contain; no sensor or sampling rate can recover them. Replacing the ideal sensor by a PT1000 or a DS18B20 sampled every 15 or 60 min removes only a further 0.00–0.02.

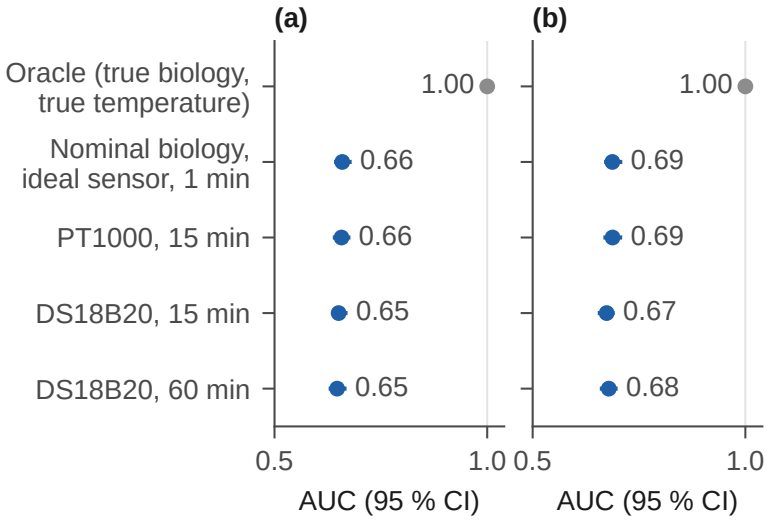


Figure 7.1 Loss budget for (a) a low-mass pouch and (b) an insulated crate (limit 3×10^4 CFU mL^{-1}, $cv = 0.10$). Grey: oracle with true biology and temperature. Blue: the growth-model score as information is removed step by step. Bars: 95 % bootstrap intervals

Table 7.1 Loss budget: mean AUC over the 12 settings of each run as information is removed, and the range of the biology gap (oracle minus ideal sensor)

Run	Oracle	Ideal	PT1000	DS18B20	DS18B20	Biology gap
		1 min	15 min	15 min	60 min	
Baseline	1.000	0.671	0.670	0.661	0.662	0.27–0.40
Extra lag	0.988	0.652	0.652	0.644	0.645	0.27–0.40
Low load	1.000	0.667	0.666	0.656	0.657	0.28–0.40
Milk flora	1.000	0.664	0.663	0.654	0.653	0.28–0.40
Listeria	0.950	0.738	0.737	0.725	0.726	0.09–0.34

7.2 Against simple rules (RQ3)

On identical trips, the growth-model score beats time above 5 °C in 12 of 12 settings and the maximum temperature in 12 (Table 7.2, Fig. 7.2). Against degree-hours above 4 °C, a rule that needs no growth model, the gain is positive with an interval excluding zero in 10 of 12 settings, but small: 0.00–0.02 AUC in the baseline run and up to 0.04 when extra lag is present. This is the behaviour predicted in Chapter 3: degree-hours are the linear part of the same integral, and the growth model gains only on the large excursions where the quadratic part matters. Time above 8 °C and the maximum temperature discard the duration or the depth of an excursion and rank trips worst.

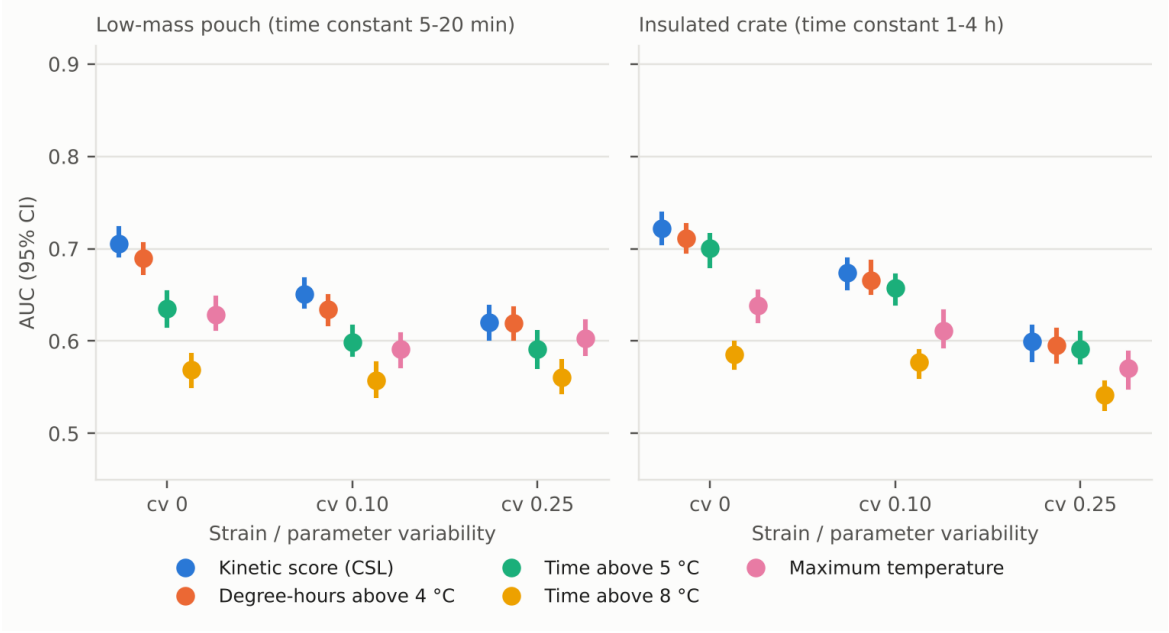


Figure 7.2 AUC of the five scores from a DS18B20 record sampled every 15 min in the baseline run, by product and strain variability (limit 3×10^4 CFU mL^{-1}). Bars: 95 % bootstrap intervals

7.3 Sensors and sampling (RQ2)

Replacing a DS18B20 with a PT1000 gains 0.00–0.02 AUC at 15-min sampling (Table 7.3, Fig. 7.3). The reason is visible in Eq. (3.1): a sensor offset of ±0.5 °C shifts $\sqrt{\mu}$ by $0.5b$, about 4 % of its value at 4 °C, whereas the trip-to-trip spread of $T_{\min}$, h_0 and N_0 moves the rate scale and the budget by tens of per cent. Sampling hourly instead of every minute shows no detectable difference in 11 of 12 settings and at most about 0.01 where it does, and the adaptive rule is indistinguishable from fixed one-minute sampling. Excursions in the generator last at least 10 min and typically an hour; the conclusion on sampling would change if short, sharp excursions were common.

Table 7.2 AUC of the five scores from a DS18B20 record sampled every 15 min (baseline run, 4 000 trips per row). The best score in each row is in bold. The last two columns give the same kinetic score with an ideal sensor sampled every minute, and the oracle that knows each trip's biology

Limit	Product	*cv*	CSL	Degree-h > 4 °C	Time > 5 °C	Time > 8 °C	T_{max}	Ideal, 1 min	Oracle
A	Pouch	0.00	**0.705**	0.690	0.635	0.569	0.629	0.718	1.000
A	Pouch	0.10	**0.651**	0.634	0.599	0.557	0.591	0.659	1.000
A	Pouch	0.25	**0.620**	0.619	0.591	0.561	0.602	0.627	1.000
A	Crate	0.00	**0.722**	0.711	0.701	0.586	0.638	0.732	1.000
A	Crate	0.10	**0.674**	0.666	0.657	0.577	0.611	0.687	1.000
A	Crate	0.25	**0.599**	0.596	0.592	0.541	0.571	0.603	1.000
B	Pouch	0.00	**0.710**	0.686	0.636	0.564	0.587	0.723	1.000
B	Pouch	0.10	**0.656**	0.634	0.606	0.539	0.564	0.670	1.000
B	Pouch	0.25	**0.604**	0.594	0.565	0.531	0.549	0.608	1.000
B	Crate	0.00	**0.708**	0.696	0.687	0.600	0.629	0.724	1.000
B	Crate	0.10	**0.676**	0.670	0.663	0.585	0.611	0.686	1.000
B	Crate	0.25	**0.603**	0.595	0.591	0.538	0.552	0.608	1.000

Table 7.3 Paired AUC differences on identical trips: minimum/maximum over the 12 settings of each run, with the number of settings (of 12) whose 95 % paired-bootstrap interval excludes zero in brackets

Contrast	Baseline	Extra lag	Low load	Milk flora	*Listeria*
CSL − degree-h	0.001/0.023 (10)	0.006/0.037 (12)	0.001/0.023 (8)	0.004/0.025 (10)	0.002/0.023 (8)
CSL − time > 5 °C	0.008/0.074 (12)	0.001/0.104 (11)	0.005/0.066 (11)	0.006/0.075 (11)	0.010/0.077 (12)
PT1000 − DS18B20	0.003/0.015 (9)	0.003/0.013 (7)	0.005/0.018 (10)	0.002/0.015 (8)	0.005/0.020 (8)
60 min − 1 min	-0.008/0.004 (1)	-0.008/0.004 (1)	-0.011/0.008 (2)	-0.009/0.006 (0)	-0.008/0.009 (0)
Adaptive − 1 min	-0.006/0.005 (0)	-0.006/0.006 (0)	-0.008/0.009 (1)	-0.007/0.006 (0)	-0.006/0.009 (0)

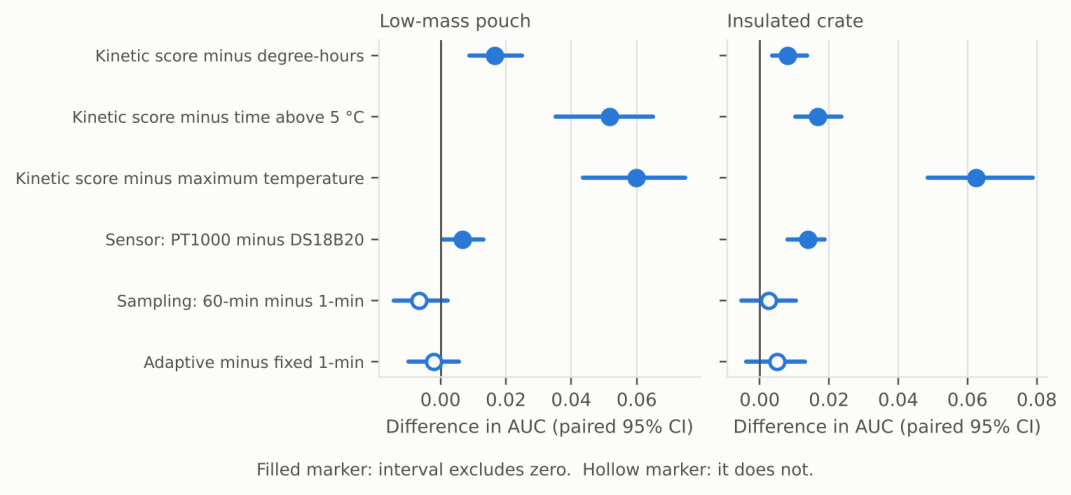


Figure 7.3 Paired AUC differences on identical trips in one baseline setting per product ($cv = 0.10$, limit 3×10^4 CFU mL^{-1}), with 95 % paired-bootstrap intervals. Filled markers: interval excludes zero

7.4 Robustness across parameter sets

Figure 7.4 summarises the four effects across all five runs. The pattern holds whether the biology is anchored to inoculated isolates, includes extra lag, starts from a realistic low load, is fitted to natural flora or describes *Listeria*: the biology gap is 0.28–0.40 (low load), 0.28–0.40 (milk flora) and 0.09–0.34 (*Listeria*); the gain over degree-hours stays below 0.04 in every run; and the sensor gain stays small.

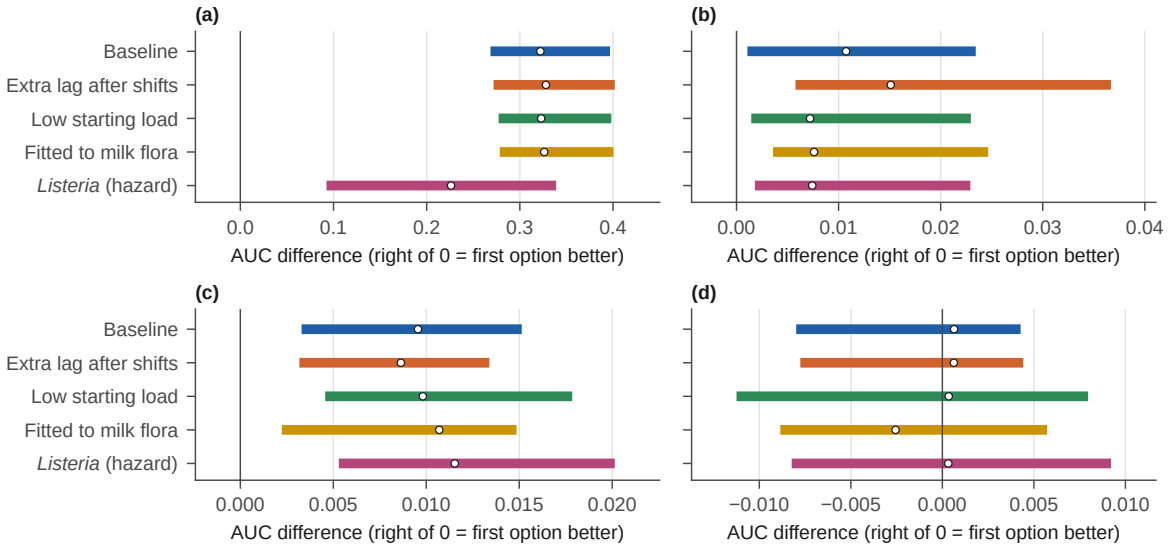


Figure 7.4 Four effects across the five parameter sets, as AUC differences on identical trips: (a) loss to unobserved biology, (b) growth-model score over degree-hours, (c) PT1000 over DS18B20, (d) hourly over one-minute sampling. Bars span the 12 settings of each run; dots mark the median. Horizontal scales differ

Table 7.4 Calibration: range of expected calibration error over the 12 settings, and mean Brier score, for each run. Lower is better

Variant	Baseline	Extra lag	Low load	Milk flora	*Listeria*
Probability, matched prior	0.03–0.07 / 0.231	0.01–0.07 / 0.236	0.02–0.11 / 0.233	0.02–0.09 / 0.234	0.08–0.41 / 0.216
Probability, overconfident	0.04–0.13 / 0.237	0.05–0.13 / 0.244	0.02–0.14 / 0.238	0.02–0.13 / 0.239	0.11–0.44 / 0.218
Single alarm	0.33–0.43 / 0.380	0.34–0.44 / 0.391	0.34–0.44 / 0.386	0.34–0.43 / 0.385	0.34–0.45 / 0.400

7.5 Calibration (RQ4)

A single alarm with nominal parameters is badly calibrated against the constructed labels (ECE 0.34–0.43; Table 7.4, Fig. 7.5). A Monte Carlo probability with a prior that matches the truth reaches ECE 0.03–0.07, and one with an overconfident prior on the growth rate 0.04–0.13. Absolute values reflect the constructed prevalence; the ordering of the three variants is the finding. A probability that admits what the log cannot know is better calibrated than a verdict that ignores it, and even a prior that understates the strain variation retains most of the benefit.

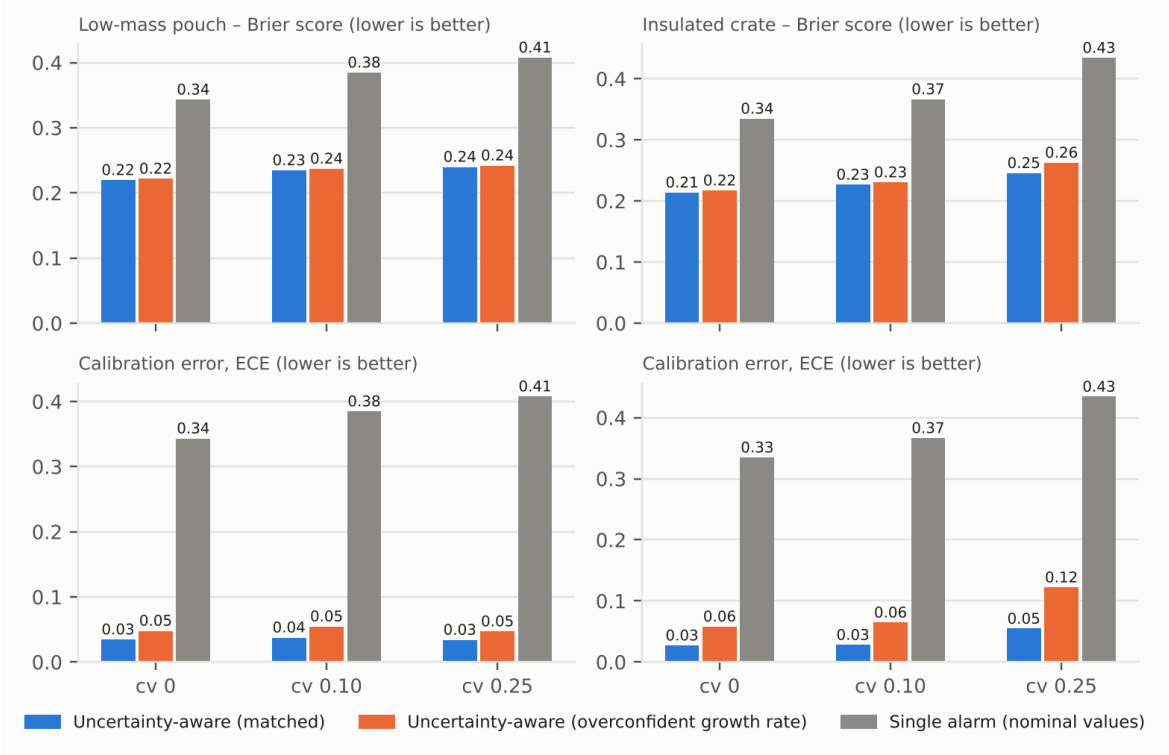


Figure 7.5 Brier score and expected calibration error of the Monte Carlo probability with a matched prior, with an overconfident prior, and of a single alarm with nominal parameters (baseline run, limit 3×10^4 CFU mL^{-1}). Lower is better

CHAPTER 8

Listeria monocytogenes: a hazard case study

Summary For a pathogen with a minimum growth temperature close to refrigeration, a warm hour costs more, but the loss budget has the same shape: biology first, instruments a distant second.

8.1 Why a pathogen behaves differently

The *Listeria* fit places $T_{\min}$ at -2.3 °C, much closer to refrigeration than the -7.7 °C of spoilage *Pseudomonas*. By Eq. (3.2), the same warm hour is therefore relatively more expensive: the rate at 13 °C is 5.9 times the rate at 4 °C, against about three times for spoilage flora. At 4 °C itself growth is slow, with a generation time of 32 h. Temperature excursions thus matter even more for the hazard than for spoilage.

8.2 Results

With labels of $+0.5$ and $+1\,\log_{10}$ growth, unobserved biology costs 0.09–0.34 AUC, a PT1000 gains 0.01–0.02, hourly sampling matches minute sampling in 12 of 12 settings, and the gain of the growth-model score over degree-hours is at most 0.02 (Tables 7.1–7.4). The lower end of the biology gap is smaller than for spoilage because the narrow starting-load range and the slow growth at 4 °C leave less room for biological differences to reorder trips.

8.3 Labels and estimator structure

The half-log label requires a caveat. The limit then lies inside the lag-to-growth transition, where the Gompertz estimator approximates the Baranyi–Roberts reference poorly; even the oracle reaches only 0.88–0.92 AUC (1.00–1.00 for the $+1$ log label). With the $+1$ log label the probability reaches ECE 0.08–0.20 against 0.34–0.41 for the alarm; with the half-log label it remains better than the alarm (0.28–0.41 against 0.42–0.45) but is poorly calibrated in absolute terms. For very small growth targets, the structure of the estimator, and not biology alone, limits accuracy.

8.4 Scope

This case study concerns growth, not illness. Risk of illness depends on dose, virulence, host susceptibility and the contamination event, none of which are modelled. The FSSAI criterion for *L. monocytogenes* in milk products is absence in 25 mL [13], which no growth threshold represents.

CHAPTER 9

Monitor design and software

Summary The results translate into a short list of design rules. The estimator runs on a microcontroller, a browser demonstrator runs the same kernel, and the whole study is reproducible from one command.

9.1 Design rules

Table 9.1 converts the results into rules for a milk cold-chain monitor.

Table 9.1 Design rules for a milk cold-chain monitor

Decision	Rule and evidence
Sensor	A ±0.5 °C digital probe suffices; a PT1000 gains at most 0.02 AUC
Sampling	15–60 min is enough unless excursions shorter than about 10 min are common
Score	Cumulative shelf life; degree-hours above 4 °C are an acceptable fallback (at most 0.04 AUC lower)
Rules to avoid	Maximum temperature and time above 8 °C; both rank trips clearly worse
Output	A probability, not an alarm (ECE 0.03–0.07 against 0.34–0.43)
Parameters	Fit to natural flora of the product; inoculated isolates overpredict by a factor of 1.62
Improvement	Measure the biology: starting load, lag and strain spread

9.2 Processing on the logger

Figure 9.1 shows how the estimator runs on a logger. A probe reading is taken on an adaptive schedule, the shelf-life rate $1/\mathrm{SL}(T)$ is read from a stored table, Eq. (3.8) is updated, and the same sum is carried for 80 sampled biological parameter sets to produce a probability. Parameter fitting and the shelf-life table are prepared once, offline. The memory required is a table of a few thousand entries and 80 running sums, well within a low-cost microcontroller.

9.3 Software

The repository is organised in layers. A kinetics module implements Eqs. (3.1)–(3.8); a simulator implements Chapter 4; the benchmark computes scores, AUCs, paired bootstraps and calibration; validation scripts reproduce Chapter 6; and scripts write every number, table and figure of the paper and of this monograph from the result files. Numerical work uses NumPy, SciPy and Matplotlib [48–50]. Fourteen automated tests (Table 9.2) run on every change. Parts of the simulation code were developed with an AI coding assistant; all of it is covered by the tests and was checked by the author.

9.4 Browser demonstrator

A public demonstrator runs the same kernel in the browser (https://copyninja07-07.github.io/dairyguard/explorer.html); a parity test ensures that it reproduces the Python implementation. A user chooses or draws a trip, such as a shop door left open or a failed truck refrigerator, and sees the probability that the milk has passed the limit, the simulated fate of 60 cartons and the comparison with a single alarm. The demonstrator is software only; a hardware logger is future work.

Example values under each block: 40 min after a truck door was left open. Steps 2–5 run on the logger.

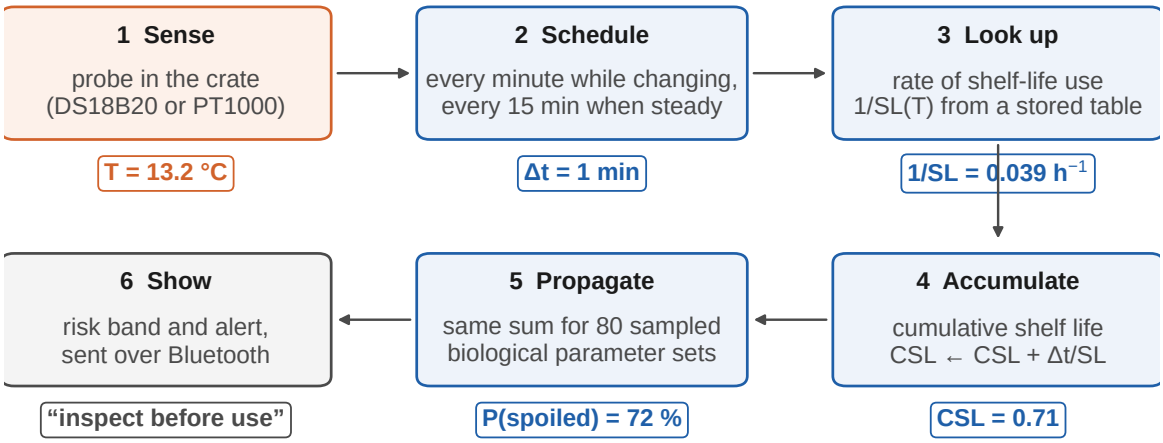


Figure 9.1 Processing on the logger, with example values 40 min after a truck door was left open. Steps 2–5 run on the device; parameter fitting and the shelf-life table are prepared offline

Table 9.2 Automated tests

Group	What is checked
Kinetics (4)	Gompertz agrees with exponential growth; reference and estimator agree at constant temperature with equal parameters; the look-up table matches the closed form; shelf life falls with temperature
Validation (3)	Fitted parameters are plausible; the constant-4 °C and two-excursion times of Lin et al. are reproduced within 10 %
Statistics (6)	Fast AUC matches a reference implementation; known AUC values are recovered; the paired difference of identical scores is zero; a real improvement is detected; the probability rises with temperature; calibration of perfect and poor forecasts
Parity (1)	The browser kernel matches the Python kernel

CHAPTER 10

Discussion

10.1 Where effort belongs

In every run, what the thermometer cannot observe costs about an order of magnitude more accuracy than any choice of sensor or sampling interval. Equation (3.9) explains why: a log measures the integral of the growth rate well, but the rate scale and the budget against which the integral is compared vary from carton to carton. A more accurate thermometer is therefore a poor way to improve a spoilage estimate, and a low-cost probe sampled every 15–60 min loses little under the assumptions made here. Effort is better spent on the biology: measuring the starting load or a proxy of handling history, learning the lag and strain spread of the product at hand, and reporting a probability instead of a verdict.

10.2 Consistency with earlier studies

Sensitivity analyses of milk spoilage rank storage temperature among the most influential inputs [35–38]; this study finds that biology dominates. The two answer different questions. When temperature is drawn from a distribution, it drives the spread of outcomes; once it is measured, the residual uncertainty is biological. The small gain of the growth-model score over degree-hours agrees with the long practical use of degree-hour and time–temperature integrators [9, 22], and the finding that parameters from inoculated isolates overpredict natural flora agrees with the practice of choosing a fast isolate as a worst case.

10.3 Hazard and spoilage

For spoilage flora a warm hour is costly but the baseline rate is already high; for *Listeria*, growth at 4 °C is slow but rises about 5.9-fold by 13 °C. The same benchmark and the same design rules hold for both, which supports treating temperature excursions as the common risk factor while recognising that a growth model says nothing about illness.

10.4 Limitations

(i) The reference and the estimator both reduce the temperature history to an integral of the growth rate, so the benchmark measures how much accuracy is lost to the listed sources, not whether a growth-model score describes growth better than another model; the structure-risk run relaxes one structural assumption only.

(ii) The strain spread, the starting load, the excursion statistics and the *Listeria* lag are assumed (Appendix A).

(iii) Labels are constructed, with prevalence near 0.5 for spoilage and 0.08–0.38 for *Listeria*.

(iv) ComBase records come from heterogeneous studies and report maximum rates, not curves, so lag and responses to temperature shifts are untested on them.

(v) A single experiment validates growth under fluctuating temperature, and no new plate counts were made.

(vi) Neither organism class represents spore-formers, which account for a large share of milk spoilage [10].

(vii) For very small growth targets, such as the half-log *Listeria* label, the estimator's structure rather than biology limits accuracy.

CHAPTER 11

Conclusions

A temperature log can say much about how much growth the milk *could* have had, and little about how much it *did*. In an open benchmark of 240 000 trips anchored to published and ComBase data, unobserved biology costs 0.27–0.40 AUC; a degree-hour count comes within 0.04 of a growth model; sensor quality and sampling interval matter little; and a Monte Carlo probability is far better calibrated than an alarm. The answers to the four research questions are therefore:

RQ1. Biology the log cannot see is the dominant loss in every run.
RQ2. A better sensor or faster sampling recovers at most about 0.02 AUC.
RQ3. A growth-model score clearly beats threshold and maximum rules but only narrowly beats degree-hours.
RQ4. A probability is far better calibrated than a single alarm.

Four directions follow: measurements of real logger data from Indian dairy routes to replace the assumed excursion model; challenge tests with natural flora under fluctuating temperature, with new plate counts; methods that measure the biology of a lot cheaply at dispatch; and a hardware logger running the estimator. An extended study, which develops the kinetic theory behind these findings, is in preparation.

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APPENDIX A

Parameters and provenance

Every parameter of the simulation is listed in Table A.1 with its value and its source. Values marked as assumed are the main candidates for measurement.

Table A.1 Model parameters, their values and their provenance, as recorded in `data/params.csv`. SOURCED: taken from a cited study; FITTED: estimated from data in this work; ASSUMED: no direct source

Parameter	Value	Unit	Provenance
Tmin_mean	-7.7	degC	SOURCED: Neumeyer 1995 thesis Table 4.9 (-7.7 C, 68 growth curves, broth and milk). Literature mean 265.4 K = -7.75 C. Lin 2016 own data give -9.15 C
Tmin_sd	1.5	degC	ASSUMED. Neumeyer Table 5.1: ±0.7 K is the 95% CI of the MEAN over 64 sources; their SD is 2.83 K but includes method noise. 1.5 K lies between the two and should be replaced by a measured strain SD
b_nominal	0.030	sqrt(1/h)/degC (ln basis)	SOURCED, two independent sources agree: Neumeyer 1995 Table 4.9 (b=0.00465 on sqrt(1/generation time in min)) converts to 0.030; Lin et al. 2016 (growth rates 0.056/0.17/0.46 per h at 4/15/29 C, log10 basis) gives 0.027
b_cv	0 / 0.10 / 0.25	fraction	ASSUMED sensitivity axis, not measured
h0_truth	U(2.5,5.0)	dimensionless	SOURCED RANGE from Lin 2016 Table 2: lag x growth rate (ln basis) = 3.8, 5.0 and 3.0 at 4, 15, 29 C. Inoculated UHT milk, so natural contamination may differ
h0_estimator	3.8	dimensionless	SOURCED: Lin 2016 at 4 C (lag 29.5 h)
log10N0_truth	U(1.5,2.0)	log10 CFU/mL	ASSUMED. Martin et al. 2021 notes post-pasteurisation contamination is often very low (1 to 50 cells per 100 mL), so this is probably high and needs measuring
log10N0_estimator	1.7	log10 CFU/mL	ASSUMED
log10Nmax	9.0	log10 CFU/mL	ASSUMED (typical order for milk psychrotrophs)
Nlimit_A	3e4	CFU/mL	SOURCED as a number: FSSAI Appendix B v2 (04.11.2024) Table 2A, pasteurized milk APC m=3x10^4/mL. A process-hygiene criterion used here as a quality marker. US PMO limit 20,000 CFU/mL (Martin 2021)
Nlimit_B	1e6	CFU/mL	SUPPORTED: about 10^6 CFU/mL is where quality deteriorates (Martin et al. 2021, citing Ranieri & Boor 2009; Lin 2016 citing Momani & Natsheh 2012)
label_shelf_life	83 (A) / 113 (B)	h	DESIGN: time for the estimator at nominal parameters held at 4 C to reach the limit; set for label balance before any method comparison
events_per_trip	Poisson, 1 per 144 h of horizon, max 8	count	ASSUMED
event_peak_ambient	12 / 14.3 / 17.5 / 30.7 (w .30/.25/.30/.15)	degC	ASSUMED: representative excursion peaks for transport and retail handling
event_duration	lognormal median 60 min, sigma 0.8, clip 10-480	min	ASSUMED
product_time_constant	low-mass U(5,20) min; insulated U(1,4) h	h	ASSUMED; to be measured on real packs
base_temp	U(2.5,5.0)	degC	ASSUMED
shift_lag_fraction	0 (baseline) / 0.25 (structure-risk run)	fraction of h0	SOURCED VALUE, ASSUMED IMPLEMENTATION: Zwietering et al. 1994 (additional lag is one quarter of the lag at the new temperature, Lactobacillus plantarum); Lin 2016 saw extra lag after shifts in milk
sensor_DS18B20	bias U(-0.5,0.5), noise sd 0.05, res 0.0625	degC	accuracy ±0.5 C from the manufacturer datasheet; noise ASSUMED

Parameter	Value	Unit	Provenance
sensor_PT1000_ MAX31865	bias U(-0.15,0.15), noise sd 0.03, res 0.03125	degC	accuracy ±0.15 C (IEC 60751 Class A tolerance at 0 C); noise ASSUMED
combase_fit	b 0.0256, Tmin -6.6 C (n=50 robust); log resid sd 0.16		EMPIRICAL: 51 ComBase records (Tasmanian Institute of Agriculture, pasteurised milk, 2.7-14.9 C); one record with an implausible temperature removed; Neumeyer/Lin parameters overpredict these by 1.6x
runs_reported	baseline; extra lag; low starting load; ComBase-fitted milk flora; Listeria		all use seeds fixed in src/benchmark_v2.py; 4000 trips per setting
lis_b	0.0235	ln basis	FITTED: 46 ComBase records of Listeria monocytogenes in milk, conditions not specified, 2-16 C; bootstrap 95% 0.0203-0.0265. Held-out check on 53 records: bias factor 1.02, accuracy factor 1.37 (split by condition, not by study)
lis_Tmin	-2.3	degC	FITTED: same fit; bootstrap 95% -4.2 to -0.9 C
lis_Tmin_sd	1.0	K	ASSUMED: of the order of the bootstrap standard error (about 0.8 K)
lis_h0	3.0 (estimator); U(2.0,4.5) (truth)	dimensionless	ASSUMED: no Listeria lag data were used; range borrowed from Pseudomonas
lis_N0	U(-1.25,-0.75) narrow; estimator -1.0	log10 CFU/mL	ASSUMED: about 0.06-0.18 cells per mL. Kept narrow on purpose: the +0.5 log label sits at -0.5, so a wider start range would put some trips above the label at time zero
lis_labels	+0.5 log10 (A) and +1 log10 (B) growth over starting load; horizons 100 h and 202 h		ASSUMED round-number labels, NOT regulatory limits (FSSAI Table 2B requires absence in 25 mL)

APPENDIX B

Data

Table B.1 lists the ComBase records of spoilage flora used in Chapter 6. The complete compilation of 373 milk records, including the 102 *Listeria* records with their conditions and sources, is in the repository (`data/combase_milk_records.csv`).

Table B.1 ComBase records of aerobic spoilage bacteria in pasteurised milk (Tasmanian Institute of Agriculture; 51 records). μ_{max}: maximum growth rate in decimal-logarithm units per hour

T/°C	μ_{max}/h^{-1}	T/°C	μ_{max}/h^{-1}	T/°C	μ_{max}/h^{-1}
2.7	0.0171	6.8	0.0242	10.7	0.0866
3.4	0.0139	7.3	0.0613	11.1	0.0910
3.4	0.0264	7.4	0.0551	11.5	0.0852
4.2	0.0334	8.1	0.0605	11.5	0.1040
4.2	0.0313	8.2	0.0653	11.6	0.0948
4.3	0.0426	8.7	0.0615	12.1	0.0961
4.6	0.0391	8.8	0.0706	12.5	0.1070
4.9	0.0360	8.8	0.0676	12.6	0.0750
5.0	0.0396	9.3	0.0623	12.6	0.1270
5.2	0.0489	9.4	0.0826	12.7	0.1040
5.7	0.0492	9.6	0.0749	13.2	0.1330
5.8	0.0520	9.6	0.0658	13.3	0.1110
5.9	0.0431	9.9	0.0859	13.8	0.1230
5.9	0.0519	10.1	0.0804	13.9	0.0910
6.3	0.0512	10.2	0.0715	14.2	0.1310
6.6	0.0486	10.5	0.0813	14.8	0.1500
6.6	0.0562	10.7	0.0740	14.9	0.1200

APPENDIX C

Reproducing the results

All results are generated from fixed random seeds. With Python 3.10 or later, the packages in `requirements.txt` and a LaTeX installation:

Command	Effect
`make test`	run the 14 automated tests
`make runs`	run the five benchmark runs (about three hours on two cores)
`make validate`	reproduce the validation of Chapter 6
`make paper`	write all numbers, tables and figures and build the paper
`make book`	build this monograph
`make all`	everything above

Every number quoted in the text is read from the result files by a script and written into the document as a macro; none is typed by hand.

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